+852-94965486
cd.melvin.wang@outlook.com
GitHub Profile: Melvin-King
ORCID: 0009-0002-2164-406X
LinkedIn Profile

Wang "Melvin" Ming

PhD student in Computing Science
University of Alberta, Canada
Personal Website: https://melvin-king.github.io

Education

•University of Alberta *2026-Present*
PhD in Computing Sciene Ongoing

•The Chinese University of Hong Kong, Hong Kong *2025-2026*
Master of Science in Information Engineering Graduated

•University of Cambridge *2022*
Mathematics for Engineers Programme Certificate Granted

•The Hong Kong Polytechnic University, Hong Kong *2020-2024*
Bachelor of Science in Computer Science Graduated

Experience

•Department of Electrical and Computer Engineering, University of Waterloo *January 2026 - April 2026*
Co-researcher Canada
- Exploring the cost structure of MLLM educational deployment in block-based programming.

•ARISE Lab, Department of Computer Science and Engineering, CUHK *March 2025 - September 2025*
Co-researcher Hong Kong
- Front-end Coding intelligence analysis for Multimodal LLMs Baseline
- Data Collection and Processing, Metrics, Evaluation, Benchmarking
- Building AI Agent system as Metareviewer for an academic paper review
- Agentic AI, MLLM, RAG, MoE

•Department of Computing, PolyU *June 2024 - September 2024*
Research Assistant (Full-time) Hong Kong
- Temporal Graph Neural Network (TGNN), Link Prediction and Node Classification
- Model Testing, Evaluation Baseline testing and Benchmarking

•Code for Asia Ltd. *June-August, 2023*
Student internship Singapore
- Front-end Development
- AI Problem-solving Architect

•Department of Building and Real Estate, PolyU *April 2023- June 2024*
Research Assistant (Part-time) Hong Kong
- LLM-driven Online Social Network Analysis
- Data Retrieval, Large Language Model, Social Network Analysis, Paper Completion

Publications & Preprints

•J. Xiao, M. Wang, M. H. Lam, Y. Wan, J. Liu, Y. Huo and M. R. Lyu, "DesignBench: A Comprehensive Benchmark for MLLM-based Front-end Code Generation", in arXiv preprint, 2025, doi: 10.48550/arXiv.2506.06251
Keywords: MLLM, Code Inteligence, Software Engineering, Benchmark

•M. Wang, R. Ma, G. Q. Shen and J. Xue, "How Large Language Model Empowers the Analysis of Online Public Engagement for Mega Infrastructure Projects: cases in Hong Kong", in IEEE Transactions in Engineering Management (IEEE-TEM), 2025, doi: 10.1109/TEM.2025.3553595
Keywords: Large Language Model, Social Network Analysis, Public Engagement, Mega Projects

•R. Ma, G. Q. Shen, P. Lou and M. Wang, "Large Language Model-based data-driven framework for digital transformation in construction industry," in ASCE International Conference on Computing in Civil Engineering 2024 (i3CE), doi: 10.1061/9780784486115.012
Keywords: Large Language Model, Digital Transformation, Engineering Management

Projects

- **Data vs Pipeline Parallelism in LLM Training** *October 2025*

  Aims to build a VR Metaverse system for elderly rehabilitation training.

  – Implements distributed training methods, including data parallelism and pipeline parallelism across multiple GPUs.

- **Robotic-guided Self-help Rehabilitation in Metaverse for stroke patients** *March - May 2023*

  Aims to build a VR Metaverse system for elderly rehabilitation training.

  – Tools & technologies used: Leap Motion, Unity, UART for data transmission
  – The forearm wearable device is developed by PolyU BRE

- **LLM-empowered Family Healthcare Management System** *January - March 2024*

  A smarter family healthcare system with personalized medical inquiry and app operation via word inputs.

  – Tools & technologies used: Python, Java, LLM, Prompt Engineering

Technical Skills and Interests

Languages: English, Mandarin, Cantonese
Developer Tools: PyCharm, VSCode, IntelliJ IDEA
Programming Languages: C/C++, Java, Javascript, Python, R
Libraries: Torch, Tensorflow, NetworkX, Sklearn, Pandas, Matplotlib, Seaborn
Operating Systems: Windows, Linux
Areas of Interest: MLLM, Software Engineering, Agentic AI, Social Network Analysis

Positions of Responsibility

- **Reviewer,** IEEE Transactions in Engineering Management (IEEE-TEM) *May 2025-October 2025*
- **Student Ambassador,** Amazon Web Services(AWS) *2023-2024*
- **First Aid Brigade Member,** Hong Kong St. John Ambulance *July 2024-2026*

Achievements

- **Certificate of Outstanding Contribution,** AWS Student Ambassador Programme *2023*
- **Excellent Teaching Team,** STEM Education for Children, PolyU Service Learning Programme *2022*
- **3rd Prize,** National Olympics for Informatics in Provinces (NOIP), China *2018*